Prof. Kenneth M. Merz, Jr.
Editor-in-Chief, Journal of Chemical Information and Modeling

Dear Prof. Merz,

I submit the Article “Hard Constraints or Penalties? Temperature Extrapolation of MOF and COF Water Isotherms” for consideration in the Journal of Chemical Information and Modeling.

Machine-learning models of chemical data are increasingly given physical conditions, either built into the architecture or added to the training loss as penalties. Which of the two to use is usually argued, not tested. The manuscript is a pre-registered benchmark of these constraint strategies on a task that matters for water sorbents in heat pumps and water harvesters: predicting the water isotherm of a metal-organic or covalent organic framework at a temperature where none was measured.

The JCIM guidelines rightly caution against new models whose results are similar to existing methods. This paper does not propose a model that wins. It asks whether a constraint strategy changes the outcome, and it was designed so that a null answer could be told apart from an inconclusive one; here the accuracy comparison was inconclusive, and the paper says so. The comparisons, an equivalence margin, the words of every possible outcome (including a classical model that wins) and the analysis code were frozen before any confirmatory fit, and synthetic data checked beforehand that each verdict could fail. On 13 frameworks (5 MOFs, 8 COFs), in a factorial design crossing the shape constraint (none, penalty, hard) with the temperature structure, the results are:

- The hard constraints gave monotone, non-negative isotherms in every fit, whereas the penalty at its accuracy-tuned weight left shape violations in 9 of 26 extrapolations. In post hoc refits, a far heavier weight removed every violation above the counting tolerance without a clear change in the error ratio, but only a weight found on the test materials did so: the lightest weight that removed the violations on the development materials left violations in 4 of 26 test extrapolations. The hard constraint gives the guarantee without that choice; no accuracy advantage was found.
- The accuracy comparisons were inconclusive. Against the penalty model the error ratio was 1.10 (95% confidence interval 0.93 to 1.65), so the pre-registered non-inferiority claim was not shown. The characteristic curves had the lowest median errors, and a classical isostere extrapolation could not be told apart from the hard model, although its isosteres crossed in 15 of 26 extrapolations.
- The temperature dependence, left free in sign, produced heats outside 0 to 150 kJ/mol in 14 of 26 extrapolations and the worst single extrapolation, which points to constraints on the temperature dependence as the next step.

For the JCIM reader the transferable parts are the benchmark protocol, which applies to any learned model of scientific data with physical constraints, a table of every model in the design, and a short set of recommendations for learned isotherm models in the Discussion. The work also matches the journal’s data and software policy: the frozen pre-registration and its hash manifest, a public ledger of every deviation and every analysis added after the verdicts, a separately written second scorer, all run records, and a manuscript build that checks every number in the text against its source file are public (github.com/abdulhamidbatayhi123/ai-isotherm-thermo; archived at doi:10.5281/zenodo.24148253). Following the journal’s 2020 editorial on data sharing, the isotherm point lists from openly licensed sources are public, the repository’s scripts extract those of all but one other material from freely available sources, and the remaining list, from a subscription article, is available to the editor and reviewers in confidence.

The manuscript has not been published and is not under consideration elsewhere, and it has not been posted as a preprint. I have no competing interests. The use of a generative AI assistant as a tool is disclosed in the Acknowledgment and the Methods.

Sincerely,

Abdulhamid Batayhi
Department of Mechatronics Engineering, Faculty of Mechanical Engineering
Yildiz Technical University, 34220 Istanbul, Türkiye
abdulhamid.batayhi@std.yildiz.edu.tr